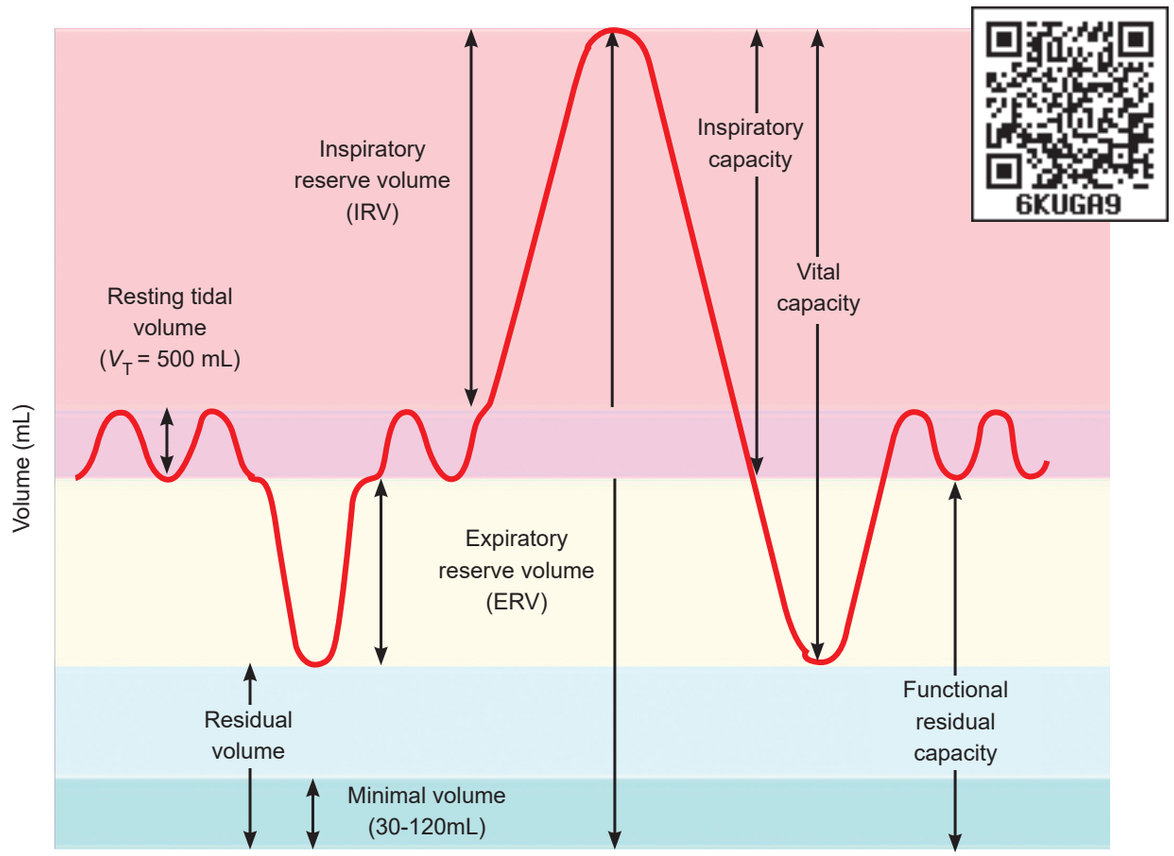


Figure 6.5 Lung volumes and capacity

tidal volume and expiratory reserve volume. EC=TV+ERV

- **Total Lung Capacity (TLC)** Thetotal volume of air which the lungs can accommodate after forced inspiration is called Total Lung Capacity. This includes the vital capacity and the residual volume. It is approximately 6000mL. TLC=VC+RV
- **Minute Respiratory Volume** The amount of air that moves into the respiratory passage per minute is called minute respiratory volume.

  Normal TV = 500mL; Normal respiratory rate = 12 times/minute

  Therefore, minute respiratory volume = 6 Litres/minute (for a normal healthy man).

Why do some people snore? – Breathing with a hoarse sound during sleep is caused by the vibration of the soft palate. Snoring is caused by a partially closed upper air way (nose and throat) which becomes too narrow for enough air to travel through the lungs. This makes the surrounding tissues to vibrate and produces the snoring sound.

Healthy lungs contain large amounts of elastic connective tissue around the alveoli, containing elastin, which makes the lung tissue elastic. People with emphysema and bronchitis have difficulty in exhaling because the enzyme elastase destroys the elastin around the alveoli and reduces the elasticity of the lungs.

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